Homework 6

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Math 23
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1 7.8

Question 1.

(a) Draw a direction field of the system of equations and sketch a few trajectories.

(b) Describe how the solutions behave as $t \to \infty$.

(c) Find the general solution of the system of equations.

$$\mathbf{x}' = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \mathbf{x}$$

Solution. The characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} 3-\lambda & -4 \\ 1 & -1-\lambda \end{pmatrix} = (3-\lambda)(-1-\lambda) + 4 = \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2 = 0.$$

So $\lambda = 1$ is a repeated eigenvalue with algebraic multiplicity 2. For $\lambda = 1$, we solve $(A - I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Since there is one zero row, the geometric multiplicity is 1, which is less than the algebraic multiplicity of 2. Therefore the matrix is defective and we need a generalized eigenvector. We solve $(A - I)\mathbf{v}^{(2)} = \mathbf{v}^{(1)}$ augmenting

$$\begin{pmatrix} 2 & -4 & 2 \\ 1 & -2 & 1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Setting $v_2 = 0$: $v_1 = 1$, so

$$\mathbf{v}^{(2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

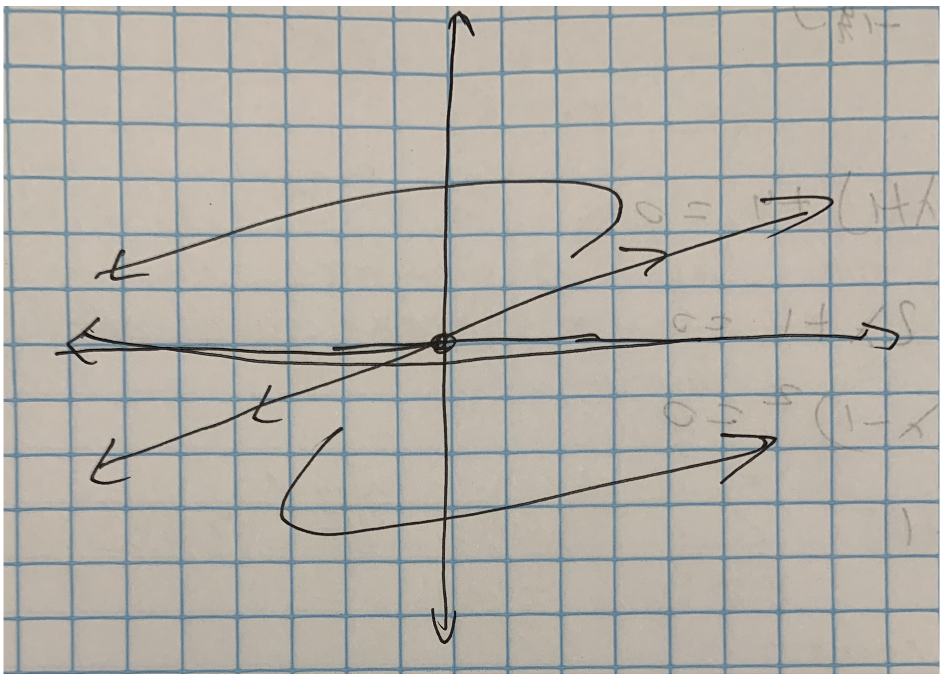
The two linearly independent solutions are

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t, \qquad \mathbf{x}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} te^t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t.$$

Therefore the general solution is

$$\boxed{\mathbf{x} = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + c_2 \left[\begin{pmatrix} 2 \\ 1 \end{pmatrix} te^t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t \right].}$$

For part (b), since $\lambda = 1 > 0$, the factor $e^t \to \infty$ as $t \to \infty$, so $\|\mathbf{x}(t)\| \to \infty$. We show the direction field for part (a) below:



$\square$

Question 5. Find the general solution of the given system of equations.

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x}$$

Solution. The characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix} = -\lambda^3 + 3\lambda + 2 = 0 \implies (\lambda - 2)(\lambda + 1)^2 = 0.$$

So $\lambda_1 = 2$ with algebraic multiplicity 1 and $\lambda_2 = -1$ with algebraic multiplicity 2. For $\lambda_1 = 2$, we solve $(A - 2I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \implies \mathbf{v}^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \implies \mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{2t}.$$

For $\lambda_2 = -1$, we solve $(A + I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The geometric multiplicity is 2. We can actually get two eigenvectors by setting one of the three variables as one and the other as negative one, hence

$$\mathbf{v}^{(2)} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \implies \mathbf{x}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{-t}.$$

and

$$\mathbf{v}^{(3)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \implies \mathbf{x}_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{-t}.$$

Therefore the general solution is

$$\boxed{\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{-t}.}$$

□

2 7.9

Question 5. Find the general solution of the given system of equations.

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2 \\ -1 \end{pmatrix} e^t$$

Solution. First we find the homogeneous solution. Considering the matrix $\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$,

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{pmatrix} = \lambda^2 - 2\lambda - 3 = 0$$

Therefore, $\lambda_1 = 3$ and $\lambda_2 = -1$. When $\lambda_1 = 3$:

$$\begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \implies \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

When $\lambda_1 = -1$:

$$\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \implies \mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Therefore,

$$\mathbf{x}_h = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}.$$

The fundamental matrix is

$$\mathbf{\Psi}(t) = \begin{pmatrix} e^{3t} & e^{-t} \\ 2e^{3t} & -2e^{-t} \end{pmatrix}.$$

We compute $\mathbf{\Psi}^{-1}(t)$. First the determinant is

$$\det(\mathbf{\Psi}) = -2e^{2t} - 2e^{2t} = -4e^{2t}.$$

Therefore,

$$\mathbf{\Psi}^{-1}(t) = \frac{1}{-4e^{2t}} \begin{pmatrix} -2e^{-t} & -e^{-t} \\ -2e^{3t} & e^{3t} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}e^{-3t} & \frac{1}{4}e^{-3t} \\ \frac{1}{2}e^{t} & -\frac{1}{4}e^{t} \end{pmatrix}.$$

Now we compute the particular solution:

$$\mathbf{\Psi}^{-1}(t)\mathbf{g}(t) = \begin{pmatrix} \frac{1}{2}e^{-3t} & \frac{1}{4}e^{-3t} \\ \frac{1}{2}e^{t} & -\frac{1}{4}e^{t} \end{pmatrix} \begin{pmatrix} 2e^t \\ -e^t \end{pmatrix} = \begin{pmatrix} e^{-2t} - \frac{1}{4}e^{-2t} \\ e^{2t} + \frac{1}{4}e^{2t} \end{pmatrix} = \begin{pmatrix} \frac{3}{4}e^{-2t} \\ \frac{5}{4}e^{2t} \end{pmatrix}.$$

$$\int \mathbf{\Psi}^{-1}(t)\mathbf{g}(t)\, dt = \begin{pmatrix} -\frac{3}{8}e^{-2t} \\ \frac{5}{8}e^{2t} \end{pmatrix}.$$

$$\begin{aligned} \mathbf{x}_p = \mathbf{\Psi}(t) \int \mathbf{\Psi}^{-1}(t)\mathbf{g}(t)\, dt &= \begin{pmatrix} e^{3t} & e^{-t} \\ 2e^{3t} & -2e^{-t} \end{pmatrix} \begin{pmatrix} -\frac{3}{8}e^{-2t} \\ \frac{5}{8}e^{2t} \end{pmatrix} \\ &= \begin{pmatrix} -\frac{3}{8}e^t + \frac{5}{8}e^t \\ -\frac{3}{4}e^t - \frac{5}{4}e^t \end{pmatrix} = \begin{pmatrix} \frac{1}{4}e^t \\ -2e^t \end{pmatrix}. \end{aligned}$$

Therefore the general solution is

$$\boxed{\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t} + \begin{pmatrix} \frac{1}{4} \\ -2 \end{pmatrix} e^t.}$$

□

3 9.1

Question 17. Consider the linear system

$$\frac{dx}{dt} = a_{11}x + a_{12}y, \qquad \frac{dy}{dt} = a_{21}x + a_{22}y,$$

where a_{11}, a_{12}, a_{21}, and a_{22} are real-valued constants. Let $p = a_{11} + a_{22}$, $q = a_{11}a_{22} - a_{12}a_{21}$, and $\Delta = p^2 - 4q$. Observe that p and q are the trace and determinant, respectively, of the coefficient matrix of the given system. Show that the critical point $(0, 0)$ is a

(a) Node if $q > 0$ and $\Delta \geq 0$;

(b) Saddle point if $q < 0$;

(c) Spiral point if $p \neq 0$ and $\Delta < 0$;

(d) Center if $p = 0$ and $q > 0$.

Proof. We rewrite the system in standard form $\mathbf{x}' = A\mathbf{x}$, then

$$\det(A - \lambda I) = \begin{pmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{pmatrix} = (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0$$

$$\lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) = 0$$

$$\lambda^2 - p\lambda + q = 0.$$

By the quadratic formula,

$$r_{1,2} = \frac{p \pm \sqrt{p^2 - 4q}}{2} = \frac{p \pm \sqrt{\Delta}}{2}.$$

Part (a): Node if $q > 0$ and $\Delta \geq 0$. Since $\Delta \geq 0$, the eigenvalues are real

$$r_{1,2} = \frac{p \pm \sqrt{\Delta}}{2}.$$

Since $q = r_1 r_2 > 0$ (Vieta's Theorem), both eigenvalues have the same sign. Since $\Delta \geq 0$, they are real and both nonzero. Therefore the origin is a node.

Part (b): Saddle point if $q < 0$. Since $q = r_1 r_2 < 0$, the eigenvalues must have opposite signs. Since $\Delta = p^2 - 4q > p^2 > 0$ when $q < 0$, the eigenvalues are real and distinct with opposite signs. Therefore the origin is a saddle point.

Part (c): Spiral point if $p \neq 0$ and $\Delta < 0$. Since $\Delta < 0$, the eigenvalues are complex conjugates

$$r_{1,2} = \frac{p}{2} \pm i\frac{\sqrt{|\Delta|}}{2}.$$

The real part is $\frac{p}{2} \neq 0$ since $p \neq 0$. Therefore the origin is a spiral point.

Part (d): Center if $p = 0$ and $q > 0$. Since $p = 0$ and $\Delta = p^2 - 4q = -4q < 0$ (because $q > 0$), the eigenvalues are purely imaginary

$$r_{1,2} = \pm i\frac{\sqrt{4q}}{2} = \pm i\sqrt{q}.$$

Since the real part is zero, the origin is a center. □

4 9.2

Question 14. Find an equation of the form $H(x, y) = c$ satisfied by the trajectories.

$$\frac{dx}{dt} = 2y, \qquad \frac{dy}{dt} = 8x$$

Solution. First, dividing gives

$$\frac{dy}{dx} = \frac{8x}{2y} = \frac{4x}{y}$$
$$y\,dy = 4x\,dx$$
$$\frac{y^2}{2} = 2x^2 + C.$$

Therefore,

$$\boxed{H(x, y) = y^2 - 4x^2 = c.}$$

□

Question 17. Find an equation of the form $H(x, y) = c$ satisfied by the

$$\frac{dx}{dt} = -x + y, \qquad \frac{dy}{dt} = -x - y$$

Solution. First, dividing gives

$$\frac{dy}{dx} = \frac{-x - y}{-x + y}.$$

Since the right side is not separable, we use the substitution $v = \dfrac{y}{x}$, so $y = vx$ and $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. Substituting gives

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{1+v}{1-v} \\ x\frac{dv}{dx} &= \frac{1+v^2}{1-v} \\ \frac{1-v}{1+v^2}\,dv = \left[\frac{1}{1+v^2} - \frac{v}{1+v^2}\right] dv &= \frac{dx}{x} \end{aligned}$$

Integrating both sides gives

$$\arctan(v) - \frac{1}{2}\ln(1+v^2) = \ln|x| + C.$$

Substituting back $v = \dfrac{y}{x}$ simplifies to

$$\boxed{H(x, y) = \arctan\left(\frac{y}{x}\right) - \frac{1}{2}\ln(x^2 + y^2) = c.}$$

□

Question 19. Find an equation of the form $H(x, y) = c$ satisfied by the trajectories.

$$\frac{dx}{dt} = y, \qquad \frac{dy}{dt} = -\sin x \qquad \text{(undamped pendulum)}$$

Solution. Solution. First, dividing gives

$$\frac{dy}{dx} = \frac{-\sin x}{y}.$$

Separating variables:

$$y\,dy = -\sin x\,dx.$$

Integrating both sides:

$$\frac{y^2}{2} = \cos x + C.$$

Therefore,

$$\boxed{H(x, y) = \frac{y^2}{2} - \cos x = c.}$$

□

□

5 9.3

Question 2. Verify that $(0, 0)$ is a critical point, show that the system is locally linear, and discuss the type and stability of the critical point $(0, 0)$ by examining the corresponding linear system.

$$\frac{dx}{dt} = (1 + x)\sin y, \qquad \frac{dy}{dt} = 1 - x - \cos y$$

Solution. Substituting $x = 0$, $y = 0$ gives

$$(1 + 0)\sin(0) = 0, \qquad 1 - 0 - \cos(0) = 1 - 1 = 0,$$

so the origin is a critical point. Since $(1+x)\sin y$ and $1-x-\cos y$ are products and sums of sin, cos, and polynomials, all second partial derivatives exist and are continuous everywhere. Therefore the system is locally linear. Lastly, we write the system as $\mathbf{x}' = A\mathbf{x} + \mathbf{g}(\mathbf{x})$ where $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so

$$g = \begin{pmatrix} (1 + x)\sin y - y \\ 1 - \cos y \end{pmatrix}.$$

The eigenvalues of A are

$$\det(A - \lambda I) = \lambda^2 + 1 = 0 \implies \lambda = \pm i.$$

Since the eigenvalues are purely imaginary, the linearization is indeterminate about the nonlinear system, even if in the linear system it is a center and stable; the origin could be a center or a spiral point. $\square$

Question 8.

$$\frac{dx}{dt} = x + x^2 + y^2, \qquad \frac{dy}{dt} = y - xy$$

(a) Determine all critical points of the given system of equations.

(b) Find the corresponding linear system near each critical point.

(c) Find the eigenvalues of each linear system. What conclusions can you then draw about the nonlinear system?

(d) Draw a phase portrait of the nonlinear system to confirm your conclusions, or to extend them in those cases where the linear system does not provide definite information about the nonlinear system.

Solution. Setting $x' = y' = 0$:

$$x + x^2 + y^2 = 0, \qquad y - xy = y(1-x) = 0.$$

From the second equation, $y = 0$ or $x = 1$. If $x = 1$, $2 + y^2 = 0$ has no real solution. If $y = 0$, $x + x^2 = x(1+x) = 0 \implies x = 0$ or $x = -1$. The critical points are $(0,0)$ and $(-1,0)$. Near $(0,0)$, the linear part is found by writing

$$F = x + x^2 + y^2, \qquad G = y - xy.$$

The linear part is

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues are $\lambda_1 = \lambda_2 = 1 > 0$, so the origin is an **unstable node or spiral point**. Near $(-1,0)$, we shift by setting $u = x+1$, $v = y$ so $x = u - 1$. Then,

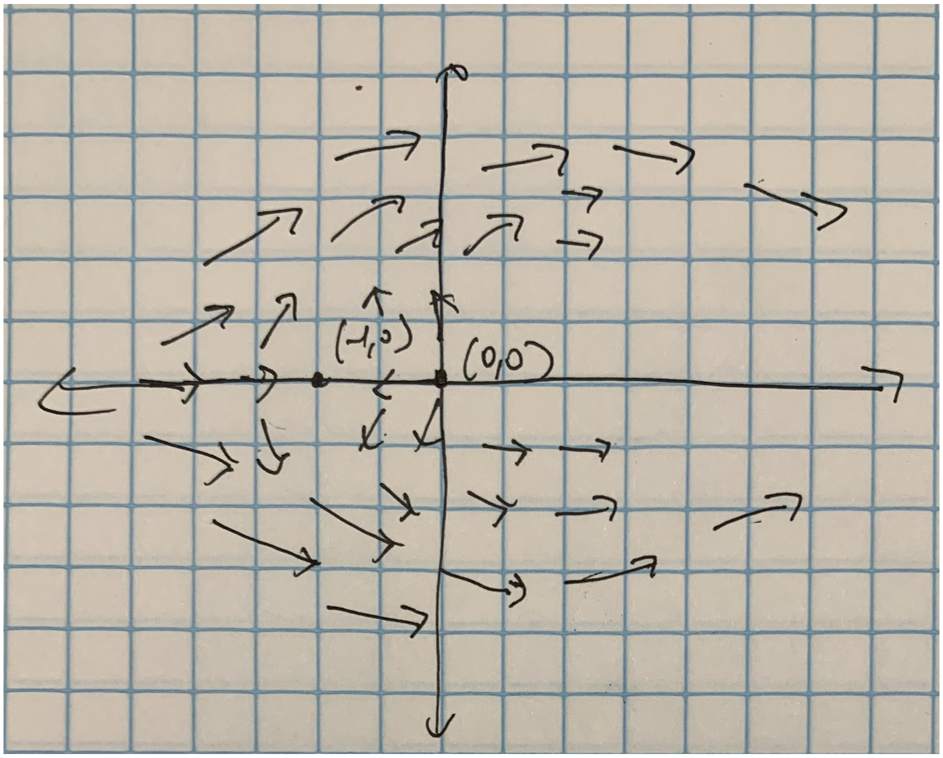
$$u' = x' = (u-1) + (u-1)^2 + v^2 = u - 1 + u^2 - 2u + 1 + v^2 = -u + u^2 + v^2$$

$$v' = y' = v - (u-1)v = v - uv + v = 2v - uv.$$

The linear part at $(u,v) = (0,0)$ is

$$B = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}.$$

The eigenvalues of B are $\lambda_1 = -1$ and $\lambda_2 = 2$, so critical point $(-1,0)$ is a **saddle point**, which is unstable. □


(-1,0)
(0,0)

Question 9.

$$\frac{dx}{dt} = (1+x)\sin y, \qquad \frac{dy}{dt} = 1 - x - \cos y$$

(a) Determine all critical points of the given system of equations.

(b) Find the corresponding linear system near each critical point.

(c) Find the eigenvalues of each linear system. What conclusions can you then draw about the nonlinear system?

(d) Draw a phase portrait of the nonlinear system to confirm your conclusions, or to extend them in those cases where the linear system does not provide definite information about the nonlinear system.

Solution. Setting $x' = y' = 0$:

$$(1+x)\sin y = 0, \qquad 1 - x - \cos y = 0.$$

From the first equation, $x = -1$ or $\sin y = 0$. If $x = -1$, substituting into the second equation gives $\cos y = 2$, which has no real solution. If $\sin y = 0$, then $y = n\pi$ for integer n. Substituting into the second equation:

$$1 - x - \cos(n\pi) = 0.$$

If n is even, $\cos(n\pi) = 1$, so $x = 0$. Critical points: $(0, 2m\pi)$. If n is odd, $\cos(n\pi) = -1$, so $x = 2$. Critical points: $(2, (2m+1)\pi)$. Therefore the critical points are

$$(0,\ 2m\pi) \quad \text{and} \quad (2,\ (2m+1)\pi), \qquad m \in \mathbb{Z}$$

Near $(0, 2m\pi)$, we set $u = x$, $v = y - 2m\pi$. Since $\sin(v + 2m\pi) = \sin v$ and $\cos(v + 2m\pi) = \cos v$, the system reduces to the same linear part as the previous problem. So,

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \implies \lambda = \pm i.$$

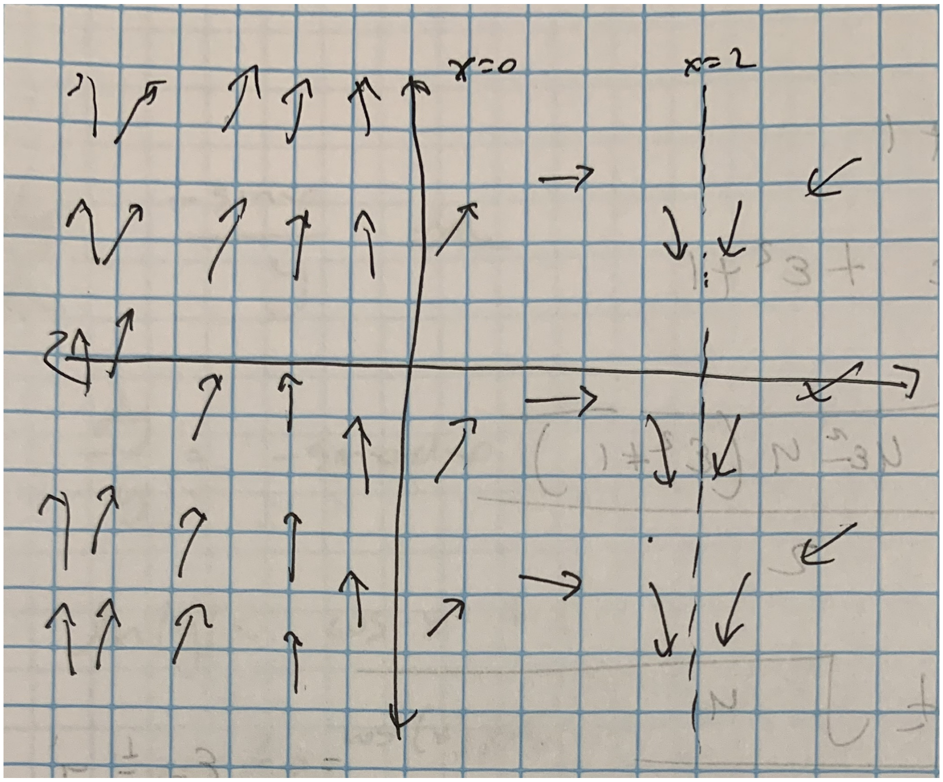
The linearization gives a **center**, but is indeterminate for the nonlinear system. Near $(2, (2m+1)\pi)$, we set $u = x - 2$, $v = y - (2m+1)\pi$. Since $\sin(v + (2m+1)\pi) = -\sin v$ and $\cos(v + (2m+1)\pi) = -\cos v$,

$$u' = -(3+u)\sin v, \qquad v' = -1 - u + \cos v.$$

The linear part at $(u, v) = (0, 0)$ is

$$B = \begin{pmatrix} 0 & -3 \\ -1 & 0 \end{pmatrix}.$$

The eigenvalues of B are $\lambda^2 - 3 = 0 \implies \lambda = \pm\sqrt{3}$, so the critical point $(2, (2m+1)\pi)$ is a **saddle point**, which is unstable. □


x=0
x=2

Question 15.

$$\frac{dx}{dt} = (1-y)(2x-y), \qquad \frac{dy}{dt} = (2+x)(x-2y)$$

(a) Determine all critical points of the given system of equations.

(b) Find the corresponding linear system near each critical point.

(c) Find the eigenvalues of each linear system. What conclusions can you then draw about the nonlinear system?

(d) Draw a phase portrait of the nonlinear system to confirm your conclusions, or to extend them in those cases where the linear system does not provide definite information about the nonlinear system.

Solution. Setting $x' = y' = 0$ gives

$$(1-y)(2x-y) = 0, \qquad (2+x)(x-2y) = 0.$$

From the first equation, $y = 1$ or $y = 2x$. From the second equation, $x = -2$ or $x = 2y$. Combining gives the critical points $(0,0)$, $(-2,1)$, $(2,1)$, and $(-2,-4)$. Near $(0,0)$, expanding $F = (1-y)(2x-y)$ and $G = (2+x)(x-2y)$:

$$F = 2x - y - 2xy + y^2, \qquad G = 2x - 4y + x^2 - 2xy.$$

The linear part is

$$A = \begin{pmatrix} 2 & -1 \\ 2 & -4 \end{pmatrix}.$$

The eigenvalues are $\lambda^2 + 2\lambda - 6 = 0 \implies \lambda = -1 \pm \sqrt{7}$, so $(0,0)$ is a **saddle point**, which is unstable. Near $(-2,1)$, we set $u = x+2$, $v = y-1$. Then,

$$u' = (-v)(2u - v - 5 + 4) = 5v + \cdots, \qquad v' = u(u - 2v - 4) = -4u + \cdots$$

Therefore,

$$B = \begin{pmatrix} 0 & 5 \\ -4 & 0 \end{pmatrix}.$$

The eigenvalues are $\lambda^2 + 20 = 0 \implies \lambda = \pm 2i\sqrt{5}$, so $(-2,1)$ is a **center** for the linearized system, which is indeterminate for the nonlinear system. Near $(2,1)$, we set $u = x-2$, $v = y-1$. With $x = u+2$, $y = v+1$, keeping only linear terms,

$$u' = (-v)(2u - v + 3) \approx -3v, \qquad v' = (u+4)(u-2v) \approx 4u - 8v.$$

The linear part is

$$C = \begin{pmatrix} 0 & -3 \\ 4 & -8 \end{pmatrix}.$$

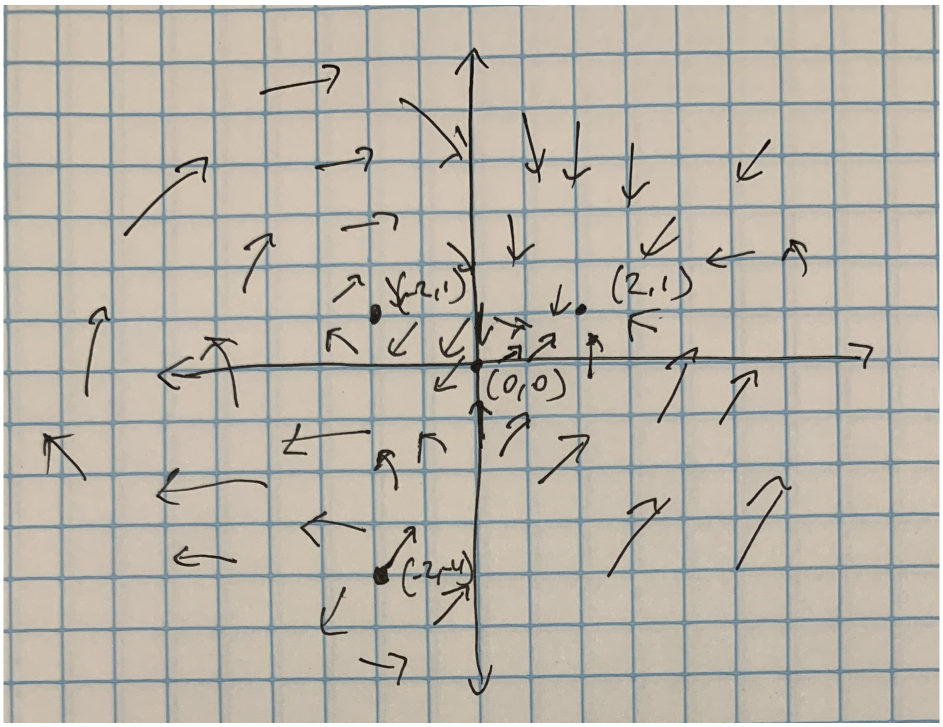
The eigenvalues are $\lambda^2 + 8\lambda + 12 = 0 \implies (\lambda+2)(\lambda+6) = 0 \implies \lambda_1 = -2,\ \lambda_2 = -6$, so $(2,1)$ is a **stable node**. Near $(-2,-4)$, we set $u = x+2$, $v = y+4$. With $x = u-2$, $y = v-4$, focusing only linear terms,

$$u' = (5-v)(2u-v) = 10u - 5v + \cdots, \qquad v' = u(u - 2v + 6) = 6u + \cdots$$

The linear part is

$$D = \begin{pmatrix} 10 & -5 \\ 6 & 0 \end{pmatrix}.$$

The eigenvalues are $\lambda^2 - 10\lambda + 30 = 0 \implies \lambda = 5 \pm i\sqrt{5}$, so $(-2,-4)$ is an **unstable spiral point**. $\square$



Question 24. In this problem we show how small changes in the coefficients of a system of linear equations can affect a critical point that is a center. Consider the system

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{x}.$$

Show that the eigenvalues are $\pm i$ so that $(0,0)$ is a center. Now consider the system

$$\mathbf{x}' = \begin{pmatrix} \epsilon & 1 \\ -1 & \epsilon \end{pmatrix} \mathbf{x},$$

where $|\epsilon|$ is arbitrarily small. Show that the eigenvalues are $\epsilon \pm i$. Thus no matter how small $|\epsilon| \neq 0$ is, the center becomes a spiral point. If $\epsilon < 0$, the spiral point is asymptotically stable; if $\epsilon > 0$, the spiral point is unstable.

Proof. For the first system

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{x},$$

the characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ -1 & -\lambda \end{pmatrix} = \lambda^2 + 1 = 0 \implies \lambda = \pm i.$$

Since the eigenvalues are purely imaginary, $(0,0)$ is a **center**. For the second system

$$\mathbf{x}' = \begin{pmatrix} \epsilon & 1 \\ -1 & \epsilon \end{pmatrix} \mathbf{x},$$

the characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} \epsilon - \lambda & 1 \\ -1 & \epsilon - \lambda \end{pmatrix} = \lambda^2 - 2\epsilon\lambda + \epsilon^2 + 1 = 0.$$

Applying the quadratic formula:

$$\lambda = \frac{2\epsilon \pm \sqrt{4\epsilon^2 - 4(\epsilon^2 + 1)}}{2} = \frac{2\epsilon \pm \sqrt{-4}}{2} = \epsilon \pm i.$$

Since $|\epsilon| \neq 0$, the real part is never zero, so the center must be a spiral point. If $\epsilon < 0$, then it is a spiral sink and the center is **asymptotically stable**. Likewise, if $\epsilon > 0$, then it is a spiral source and the point is **unstable**. $\square$

Question 25. In this problem we show how small changes in the coefficients of a system of linear equations can affect the nature of a critical point when the eigenvalues are equal. Consider the system

$$\mathbf{x}' = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \mathbf{x}.$$

Show that the eigenvalues are $r_1 = -1$, $r_2 = -1$ so that the critical point $(0,0)$ is an asymptotically stable node. Now consider the system

$$\mathbf{x}' = \begin{pmatrix} -1 & 1 \\ -\epsilon & -1 \end{pmatrix} \mathbf{x},$$

where $|\epsilon|$ is arbitrarily small. Show that if $\epsilon > 0$, then the eigenvalues are $-1 \pm i\sqrt{\epsilon}$, so that the asymptotically stable node becomes an asymptotically stable spiral point. If $\epsilon < 0$, then the roots are $-1 \pm \sqrt{|\epsilon|}$, and the critical point remains an asymptotically stable node.

Proof. For the first system

$$\mathbf{x}' = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \mathbf{x},$$

the characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} -1-\lambda & 1 \\ 0 & -1-\lambda \end{pmatrix} = (-1-\lambda)^2 = \lambda^2 + 2\lambda + 1 = 0 \implies (\lambda+1)^2 = 0.$$

So $r_1 = r_2 = -1$, and since both eigenvalues are negative, $(0,0)$ is an **asymptotically stable degenerate node**. For the second system

$$\mathbf{x}' = \begin{pmatrix} -1 & 1 \\ -\epsilon & -1 \end{pmatrix} \mathbf{x},$$

the characteristic equation is

$$\det(A - \lambda I) = \det \begin{pmatrix} -1-\lambda & 1 \\ -\epsilon & -1-\lambda \end{pmatrix} = \lambda^2 + 2\lambda + 1 + \epsilon = 0.$$

Applying the quadratic formula:

$$\lambda = \frac{-2 \pm \sqrt{4 - 4(1+\epsilon)}}{2} = \frac{-2 \pm \sqrt{-4\epsilon}}{2} = -1 \pm \sqrt{-\epsilon}.$$

If $\epsilon > 0$, $\sqrt{-\epsilon} = i\sqrt{\epsilon}$, so $\lambda = -1 \pm i\sqrt{\epsilon}$. The real part is $-1 < 0$, so the origin becomes an **asymptotically stable spiral point**. If $\epsilon < 0$, both roots are real. Since $0 < \sqrt{-\epsilon} < 1$ for $|\epsilon|$ small, both roots remain negative, so the origin remains an **asymptotically stable node**. $\square$